You said:

calculus hw 34:
Due Wednesday: ap calc: HW#34 pg192-194 #1-4, 6, 11-27 odd and pg.198 #74, 75, 83a (for 83 “3x” in the root, not just x)

ChatGPT said:

AP Calculus AB — Homework #34

Textbook: James Stewart, *Calculus*, 8th Edition
Due: Wednesday, October 21, 2026

Pages	Assigned problems	Count
192–194	1, 2, 3, 4, 6	5
192–194	11, 13, 15, 17, 19, 21, 23, 25, 27	9
198	74, 75, 83(a)	3
Total		17

Important correction: For problem 83, the expression under the square root contains **3x**, not just x.

This appears to be from Stewart's Chapter 3, Section 3.10, **Linear Approximations and Differentials**, but I'd like to verify the exact exercises.

We can review Jerry's homework the same way as before: first establish the exact problem statement, then work through the solution, including the reasoning, formulas, and common mistakes.

Shall we start with problem #1 on page 192? You can upload a photo of the textbook page so I can reproduce the correct problem rather than guess its wording.

You said:

show me problem 1 statement and solution

ChatGPT said:

Problem 1 — Page 192

Stewart Calculus, 8th Edition, Section 2.9

Problem statement: Find the linearization $L(x)$ of the function at a.

$$f(x) = x^3 - x^2 + 3, \qquad a = -2$$

Verified against the textbook solution reference. Calculus (8th ...

Step 1: Recall the linearization formula

$$\boxed{L(x) = f(a) + f'(a)(x - a)}$$

This is the equation of the tangent line at $x = a$.

Step 2: Calculate $f(a)$

$$f(-2) = (-2)^3 - (-2)^2 + 3$$

$$= -8 - 4 + 3 = \boxed{-9}$$

Step 3: Calculate the derivative

$$f'(x) = 3x^2 - 2x$$

At $a = -2$:

$$f'(-2) = 3(-2)^2 - 2(-2)$$

$$= 12 + 4 = \boxed{16}$$

Step 4: Write the tangent line in point-slope form

$$L(x) - (-9) = 16(x - (-2))$$

$$\boxed{L(x) + 9 = 16(x + 2)}$$

Or, simplified:

$$\boxed{L(x) = 16x + 23}$$

Final answer: $L(x) = 16x + 23$.

Key concept for Jerry: Linearization is simply finding the tangent line to approximate a function near a particular point. Remember the point-slope formula $y - y_1 = m(x - x_1)$, where the slope is $f'(a)$.

You said:

could you recap all problem statements and provide solutions, in 1 single pdf doc, all 17 problems

Worked for 2m 4s

ChatGPT said:

I created one PDF covering all **17 assigned exercises** from Jerry's AP Calculus Homework #34.

Download AP Calculus HW #34 — Worked Solution Guide (PDF) ↗

The three-page document includes paraphrased problem statements, step-by-step solutions where the exercise details could be verified, and the corrected cube-root expression for #83(a).

Important: This is not yet a fully verified solution set. Problems **#6, #19, and #21** still need the textbook questions, and #4 has a discrepancy in an online solution that I've flagged.

Please upload photos of pages 192–194 and 198. With those, I can verify the questions and complete the remaining solutions accurately rather than risk Jerry studying incorrect exercises.

Jerry_AP_Calculus_HW34_Worked_Guide.pdf
PDF

You said:

problem 2 statement and solution

ChatGPT said:

Problem 2 — Page 192

Stewart Calculus, 8th Edition — Section 2.9: Linear Approximations and Differentials

Problem statement

Find the linearization $L(x)$ of the function at a.

$$\boxed{f(x) = \sin x, \qquad a = \frac{\pi}{6}}$$

Step 1: Recall the linearization formula

$$\boxed{L(x) = f(a) + f'(a)(x - a)}$$

Step 2: Find $f(a)$

$$f\left(\frac{\pi}{6}\right) = \sin\frac{\pi}{6} = \frac{1}{2}$$

Step 3: Find $f'(a)$

Differentiate:

$$f'(x) = \cos x$$

Evaluate at $a = \frac{\pi}{6}$:

$$f'\left(\frac{\pi}{6}\right) = \cos\frac{\pi}{6} = \frac{\sqrt{3}}{2}$$

Step 4: Substitute into the formula

$$L(x) = f(a) + f'(a)(x - a)$$

$$\boxed{L(x) = \frac{1}{2} + \frac{\sqrt{3}}{2}\left(x - \frac{\pi}{6}\right)}$$

Final answer (point-slope form):

$$\boxed{L(x) - \frac{1}{2} = \frac{\sqrt{3}}{2}\left(x - \frac{\pi}{6}\right)}$$

Key concept: Linearization is the tangent-line equation. The point is $\left(\frac{\pi}{6}, \frac{1}{2}\right)$, and the slope is $\frac{\sqrt{3}}{2}$.

Note: This statement is taken from the previously prepared homework guide, not independently confirmed against a photograph of the textbook page.

You said:

problem 3 statement and solution

ChatGPT said:

Problem 3 — Page 192

Stewart Calculus, 8th Edition — Section 2.9: Linear Approximations and Differentials

Problem statement

Find the linearization $L(x)$ of the function at a.

$$\boxed{f(x) = \sqrt{x}, \qquad a = 4}$$

Step 1: Recall the linearization formula

$$\boxed{L(x) = f(a) + f'(a)(x - a)}$$

Step 2: Calculate $f(a)$

$$f(4) = \sqrt{4} = \boxed{2}$$

Step 3: Find the derivative

Rewrite the function:

$$f(x) = x^{1/2}$$

Using the power rule:

$$f'(x) = \frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}}$$

At $a = 4$:

$$f'(4) = \frac{1}{2\sqrt{4}} = \boxed{\frac{1}{4}}$$

Step 4: Write the linearization

$$L(x) = 2 + \frac{1}{4}(x - 4)$$

Final answer in point-slope form:

$$\boxed{L(x) - 2 = \frac{1}{4}(x - 4)}$$

Or, simplified:

$$\boxed{L(x) = \frac{x}{4} + 1}$$

Key concept for Jerry: The linearization is the tangent line to $f(x)$ at $x = 4$, with slope $1/4$ and point $(4, 2)$.

Note: The problem statement comes from the previously prepared homework guide and has not been independently checked against the printed textbook page.

You said:

problem 6 statement and solution

Worked for 15s

ChatGPT said:

Problem 6 — Page 192

Stewart Calculus, 8th Edition — Section 2.9: Linear Approximations and Differentials

I found the reference page for Exercise 6 ↗, but it doesn't contain the problem statement or solution. GradeSaver

I don't want to give Jerry a solution to the wrong problem.

Please upload a photo of problem #6 on page 192. I'll provide the exact mathematical statement and a complete step-by-step solution, including the answer in point-slope form if it asks for linearization.

The relevant formula is:

$$\boxed{L(x) = f(a) + f'(a)(x - a)}$$

You said:

problem 6 on page 196

ChatGPT said:

Problem 6 — Page 196

James Stewart, Calculus, 8th Edition — Chapter 2 Review

Problem statement (paraphrased)

Identify the function f and the value a represented by the following derivative limit:

$$\boxed{\lim_{h \to 0} \frac{(2+h)^6 - 2^6}{h}}$$

Step 1: Recall the definition of a derivative

$$\boxed{f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}}$$

Step 2: Compare the expressions

We have:

$$f(a+h) = (2+h)^6$$

$$f(a) = 2^6$$

Therefore:

$$\boxed{f(x) = x^6, \qquad a = 2}$$

Step 3: Calculate the derivative (optional)

Using the power rule:

$$f'(x) = 6x^5$$

$$f'(2) = 6(2)^5 = 6(32) = 192$$

Final answer

$$\boxed{f(x) = x^6, \quad a = 2}$$

The limit itself equals:

$$\boxed{192}$$

Key concept for Jerry: Recognize that $(a+h)^n - a^n$ in a derivative limit often indicates $f(x) = x^n$.

The textbook solution reference confirms $f(x) = x^6$ and $a = 2$. GradeSaver

You said:

problem 6 in page 192-194

ChatGPT said:

Problem 6 — Page 192

Stewart Calculus, 8th Edition, Section 2.9: Linear Approximations and Differentials

Problem statement

Find the linear approximation of

$$\boxed{g(x) = \sqrt[3]{1+x}, \qquad a = 0}$$

Use it to estimate $\sqrt[3]{0.95}$ and $\sqrt[3]{1.1}$. Illustrate with a graph of the function and its tangent line.

StudyLib

Step 1: Recall the linearization formula

$$L(x) = g(a) + g'(a)(x - a)$$

Step 2: Find $g(a)$

$$g(0) = \sqrt[3]{1 + 0} = 1$$

Step 3: Find $g'(a)$

Rewrite using exponents:

$$g(x) = (1 + x)^{1/3}$$

Differentiate using the chain rule:

$$g'(x) = \frac{1}{3}(1 + x)^{-2/3}$$

At $a = 0$:

$$g'(0) = \frac{1}{3}$$

Step 4: Find the linearization

$$L(x) = 1 + \frac{1}{3}(x - 0)$$

$$\boxed{L(x) = 1 + \frac{x}{3}}$$

Step 5: Approximate the two numbers

(a) Estimate $\sqrt[3]{0.95}$

Since $1 + x = 0.95$, we have $x = -0.05$.

$$\sqrt[3]{0.95} \approx L(-0.05)$$

$$= 1 + \frac{-0.05}{3}$$

$$\boxed{\sqrt[3]{0.95} \approx 0.98333}$$

(b) Estimate $\sqrt[3]{1.1}$

Since $1 + x = 1.1$, we have $x = 0.1$.

$$\sqrt[3]{1.1} \approx L(0.1)$$

$$= 1 + \frac{0.1}{3}$$

$$\boxed{\sqrt[3]{1.1} \approx 1.03333}$$

Step 6: Graph the function and tangent line

The straight line is tangent to the cube-root function at (0, 1).

Final answers

$$\boxed{L(x) = 1 + \frac{x}{3}}$$

$$\boxed{\sqrt[3]{0.95} \approx 0.98333}$$

$$\boxed{\sqrt[3]{1.1} \approx 1.03333}$$

Key concept for Jerry: Linear approximation uses the tangent line to estimate values of a nonlinear function near the point of tangency. The closer x is to $a = 0$, the more accurate the approximation generally becomes.

You said:

recap all problem statements and solution into 1 md file

Worked for 1m 12s

ChatGPT said:

I've consolidated Jerry's **AP Calculus Homework #34** into one Markdown file.

Download HW #34 — Problem Statements & Solutions (.md) ↗

The file includes all 17 assigned problem numbers, with paraphrased statements, step-by-step calculations, final answers, and key formulas. It also includes the corrected solution for **problem #6**.

Important: Problems #19 and #21 still need their original statements, so the file contains solution methods rather than invented answers. Other reconstructed statements are marked as needing textbook verification.

Once you provide photos of pages 192–194 and 198, I can make the guide fully verified.

You said:

can you find pdf for this textbook

ChatGPT said:

Yes. I found a **free PDF download page on James Stewart's official website**, as well as the publisher's eBook.

Jerry's textbook is *Calculus, 8th Edition* by James Stewart (ISBN 9781285740621).

1. Free PDFs — Stewart's official website

Stewart Calculus 8E — PDF Downloads ↗
Free chapter PDFs and answer files

The page includes downloadable PDFs for Chapters 1–2, Chapter 3, Chapter 4, and their answers. Stewart Calcul...

Direct Chapter 1–2 PDF: Open PDF ↗

However, the website labels these files *Calculus: Early Transcendentals, 8E*. I haven't confirmed that the PDF matches Jerry's exact exercises and pagination.

2. Full textbook — official Cengage eBook

Calculus, 8th Edition — Cengage ↗

- Exact textbook ISBN: 9781285740621
- Digital access listed starting at $76.99
- Official publisher edition Cengage

3. Online textbook reading

Stewart Calculus, 8th Edition — StudyLib ↗

This site displays textbook material online, including Section 2.9, *Linear Approximations and Differentials*. Availability and rights to the uploaded copy are not verified. StudyLib

My recommendation: Start with the official Stewart PDF. For Homework #34, we specifically need pages **192–194 and 198**. Once we confirm the exercise numbering matches, I can use the textbook to check all 17 problems and prepare an accurate Markdown solution guide.

You said:

problem 4 statement and solution

ChatGPT said:

Problem 4 — Page 192

Stewart Calculus, 8th Edition, Section 2.9: Linear Approximations and Differentials

Problem statement

Find the linearization $L(x)$ of the function at a.

$$\boxed{f(x) = \frac{2}{\sqrt{x^2 - 5}}, \qquad a = 3}$$

Step 1: Recall the linearization formula

$$\boxed{L(x) = f(a) + f'(a)(x - a)}$$

Step 2: Calculate $f(a)$

$$f(3) = \frac{2}{\sqrt{3^2 - 5}} = \frac{2}{\sqrt{4}} = \boxed{1}$$

Step 3: Find the derivative

Rewrite the function:

$$f(x) = 2(x^2 - 5)^{-1/2}$$

Apply the power rule and chain rule:

$$f'(x) = 2\left(-\frac{1}{2}\right)(x^2 - 5)^{-3/2}(2x)$$

Simplify:

$$\boxed{f'(x) = -\frac{2x}{(x^2 - 5)^{3/2}}}$$

Evaluate at $a = 3$:

$$f'(3) = -\frac{2(3)}{(9 - 5)^{3/2}}$$

$$= -\frac{6}{4^{3/2}} = -\frac{6}{8}$$

$$\boxed{f'(3) = -\frac{3}{4}}$$

Step 4: Write the linearization

$$L(x) = 1 - \frac{3}{4}(x - 3)$$

Final answer (point-slope form):

$$\boxed{L(x) - 1 = -\frac{3}{4}(x - 3)}$$

Or, simplified:

$$\boxed{L(x) = \frac{13}{4} - \frac{3x}{4}}$$

Important note for Jerry

The online GradeSaver solution reports a slope of $-1/8$, but it appears to omit the derivative of the inside function $x^2 - 5$. GradeSaver

The chain rule requires multiplying by $2x$, so the correct slope for the stated function is $-3/4$.

This is exactly the kind of mistake to watch for on an AP Calculus test.

You said:

problem 6 statement and solution

ChatGPT said:

Problem 6 — Page 192

James Stewart, Calculus, 8th Edition — Section 2.9: Linear Approximations and Differentials

Problem statement

Find the linear approximation of the function

$$\boxed{g(x) = \sqrt[3]{1+x}, \qquad a = 0}$$

Use it to approximate $\sqrt[3]{0.95}$ and $\sqrt[3]{1.1}$. Illustrate by graphing the function and its tangent line.

Step 1: Recall the linearization formula

$$\boxed{L(x) = g(a) + g'(a)(x-a)}$$

Step 2: Find $g(a)$

Since $a = 0$:

$$g(0) = \sqrt[3]{1+0} = \boxed{1}$$

Step 3: Find the derivative

Rewrite the function:

$$g(x) = (1+x)^{1/3}$$

Using the power rule and chain rule:

$$g'(x) = \frac{1}{3}(1+x)^{-2/3} \cdot 1$$

At $a = 0$:

$$g'(0) = \frac{1}{3}(1)^{-2/3} = \boxed{\frac{1}{3}}$$

Step 4: Find the linearization

$$L(x) = 1 + \frac{1}{3}(x - 0)$$

$$\boxed{L(x) = 1 + \frac{x}{3}}$$

Step 5: Approximate the two values

(a) Approximate $\sqrt[3]{0.95}$

Set $1 + x = 0.95$, so $x = -0.05$.

$$\sqrt[3]{0.95} \approx L(-0.05)$$

$$= 1 + \frac{-0.05}{3}$$

$$\boxed{\sqrt[3]{0.95} \approx 0.98333}$$

(b) Approximate $\sqrt[3]{1.1}$

Set $1 + x = 1.1$, so $x = 0.1$.

$$\sqrt[3]{1.1} \approx L(0.1)$$

$$= 1 + \frac{0.1}{3}$$

$$\boxed{\sqrt[3]{1.1} \approx 1.03333}$$

Step 6: Graph the function and linear approximation

The tangent line and function meet at (0, 1). Near x = 0, the linear approximation closely follows the curve.

Final answers

$$\boxed{L(x) = 1 + \frac{x}{3}}$$

$$\boxed{\sqrt[3]{0.95} \approx 0.98333}$$

$$\boxed{\sqrt[3]{1.1} \approx 1.03333}$$

Key concept for Jerry: Linearization replaces a complicated function with its tangent line near a known point. Here, we use $g(0) = 1$ and $g'(0) = 1/3$ to estimate nearby cube roots without a calculator.

You said:

problem 11 statement and solution

Worked for 28s

ChatGPT said:

Problem 11 — Page 192–193

James Stewart, Calculus, 8th Edition — Section 2.9: Linear Approximations and Differentials

I found references to the correct textbook section, but I couldn't reliably verify the exact statement for **Exercise 11**.

Please upload a photo of problem #11 from the textbook. I'll then provide the problem statement and complete step-by-step solution, as we did for problems #4 and #6.

I want to avoid giving Jerry a solution to a different exercise or edition.

You said:

problem 11 on page 193 that has (a) and (b)

ChatGPT said:

Problem 11 — Page 193

James Stewart, Calculus, 8th Edition — Section 2.9: Linear Approximations and Differentials

Problem statement (a) and (b)

Find the differential dy of each function.

(a)

$$\boxed{y = (x^2 - 3)^{-2}}$$

(b)

$$\boxed{y = \sqrt{1 - t^4}}$$

These functions and their answers are confirmed by the textbook solution reference. GradeSaver

Solution (a)

$$y = (x^2 - 3)^{-2}$$

Step 1: Recall the differential formula

$$\boxed{dy = y'\,dx}$$

Step 2: Differentiate using the chain rule

The outer function is u^{-2}, and the inner function is $u = x^2 - 3$.

$$\frac{dy}{dx} = -2(x^2 - 3)^{-3} \cdot 2x$$

Simplify:

$$\frac{dy}{dx} = -4x(x^2 - 3)^{-3}$$

Step 3: Multiply by dx

$$dy = -4x(x^2 - 3)^{-3}\,dx$$

Final answer (a)

$$\boxed{dy = -\frac{4x}{(x^2 - 3)^3}\,dx}$$

Solution (b)

$$y = \sqrt{1 - t^4}$$

Step 1: Rewrite the square root

$$y = (1 - t^4)^{1/2}$$

Step 2: Differentiate using the chain rule

$$\frac{dy}{dt} = \frac{1}{2}(1 - t^4)^{-1/2} \cdot (-4t^3)$$

Simplify:

$$\frac{dy}{dt} = -2t^3(1 - t^4)^{-1/2}$$

$$\frac{dy}{dt} = -\frac{2t^3}{\sqrt{1 - t^4}}$$

Step 3: Multiply by dt

$$dy = -\frac{2t^3}{\sqrt{1 - t^4}}\, dt$$

Final answer (b)

$$\boxed{dy = -\frac{2t^3}{\sqrt{1 - t^4}}\, dt}$$

Key concepts for Jerry

The important distinction is between a **derivative** and a **differential**.

Derivative	Differential
$\frac{dy}{dx} = f'(x)$	$dy = f'(x)\,dx$
$\frac{dy}{dt} = f'(t)$	$dy = f'(t)\,dt$

In part (b), because the independent variable is t, the differential is dt, **not** dx.

Jerry should remember that the chain rule is necessary for both (a) and (b). The final answer must include dx or dt, because the question asks for the differential, not just the derivative.

You said:

problem 13 statement and solution

Worked for 13s

ChatGPT said:

Problem 13 — Page 193

James Stewart, Calculus, 8th Edition — Section 2.9: Linear Approximations and Differentials

Problem statement

Find the differential of each function.

(a)

$$\boxed{y = \tan\sqrt{t}}$$

(b)

$$\boxed{y = \frac{1 - v^2}{1 + v^2}}$$

These are the two functions in Exercise 13. Exercise 13

Solution (a)

$$y = \tan\sqrt{t}$$

Step 1: Recall the differential formula

$$\boxed{dy = \frac{dy}{dt}\,dt}$$

Step 2: Apply the chain rule

Recall:

$$\frac{d}{dt}\tan(u) = \sec^2(u)\cdot u'$$

Here,

$$u = \sqrt{t} = t^{1/2}$$

Therefore,

$$u' = \frac{1}{2}t^{-1/2} = \frac{1}{2\sqrt{t}}$$

Differentiate:

$$\frac{dy}{dt} = \sec^2(\sqrt{t})\cdot\frac{1}{2\sqrt{t}}$$

Step 3: Multiply by dt

$$\boxed{dy = \frac{\sec^2(\sqrt{t})}{2\sqrt{t}}\,dt}$$

Solution (b)

$$y = \frac{1 - v^2}{1 + v^2}$$

Step 1: Recall the quotient rule

$$\boxed{\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}}$$

Identify:

$$f(v) = 1 - v^2, \quad f'(v) = -2v$$

$$g(v) = 1 + v^2, \quad g'(v) = 2v$$

Step 2: Apply the quotient rule

$$\frac{dy}{dv} = \frac{(-2v)(1 + v^2) - (1 - v^2)(2v)}{(1 + v^2)^2}$$

Expand the numerator:

$$= \frac{-2v - 2v^3 - 2v + 2v^3}{(1 + v^2)^2}$$

Combine like terms:

$$\frac{dy}{dv} = \frac{-4v}{(1 + v^2)^2}$$

Step 3: Multiply by dv

$$\boxed{dy = -\frac{4v}{(1+v^2)^2}\,dv}$$

Final answers

(a)

$$\boxed{dy = \frac{\sec^2(\sqrt{t})}{2\sqrt{t}}\,dt}$$

(b)

$$\boxed{dy = -\frac{4v}{(1+v^2)^2}\,dv}$$

Key concepts for Jerry

- **Part (a): Chain rule.** Remember that $\tan\sqrt{t}$ means $\tan(\sqrt{t})$, so we must also differentiate $\sqrt{t}$.
- **Part (b): Quotient rule.** Pay particular attention to the minus sign when expanding the numerator.
- **Both parts:** The question asks for the differential dy, not just the derivative. Remember to multiply by dt or dv.

For AP Calculus, Jerry should be comfortable deriving these results using the basic differentiation rules rather than memorizing the final expressions.

You said:

problem 15 statement and solution

ChatGPT said:

Problem 15 — Page 193

James Stewart, Calculus, 8th Edition — Section 2.9: Linear Approximations and Differentials

Problem statement

Given:

$$\boxed{y = \tan x, \quad x = \frac{\pi}{4}, \quad dx = -0.1}$$

(a) Find the differential dy.

(b) Evaluate dy for the given values of x and dx. GradeSaver

Solution (a): Find the differential dy

Step 1: Recall the differential formula

$$\boxed{dy = f'(x)\,dx}$$

Step 2: Differentiate the function

Given:

$$y = \tan x$$

Recall:

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

Therefore:

$$\frac{dy}{dx} = \sec^2 x$$

Step 3: Multiply by dx

$$\boxed{dy = \sec^2 x \, dx}$$

Solution (b): Evaluate dy

We are given:

$$x = \frac{\pi}{4}, \qquad dx = -0.1$$

Step 1: Substitute into the differential

$$dy = \sec^2\left(\frac{\pi}{4}\right)(-0.1)$$

Step 2: Evaluate the trigonometric expression

Recall:

$$\sec x = \frac{1}{\cos x}$$

Since

$$\cos\frac{\pi}{4} = \frac{\sqrt{2}}{2}$$

then

$$\sec\frac{\pi}{4} = \sqrt{2}$$

Therefore:

$$\sec^2 \frac{\pi}{4} = 2$$

Step 3: Calculate

$$dy = 2(-0.1)$$

$$\boxed{dy = -0.2}$$

Final answers

(a)

$$\boxed{dy = \sec^2 x \, dx}$$

(b)

$$\boxed{dy = -0.2}$$

Key concept for Jerry

Notice that $dx = -0.1$ is negative, meaning x decreases by 0.1.

Since $\sec^2 x$ is positive at $x = \pi/4$, the differential dy is negative too.

Physical interpretation: Near $x = \pi/4$, decreasing x by 0.1 causes $y = \tan x$ to decrease by approximately 0.2.

Remember:

$$\boxed{\Delta y \approx dy = f'(x)\Delta x}$$

The differential is an approximation of the actual change in the function, not necessarily the exact change.

You said:

problem 17 statement and solution

Worked for 19s

ChatGPT said:

Problem 17 — Page 193

James Stewart, Calculus, 8th Edition — Section 2.9: Linear Approximations and Differentials

Problem statement

(a) Find the differential dy.

(b) Evaluate dy using the given values.

$$\boxed{y = \sqrt{3 + x^2}, \quad x = 1, \quad dx = -0.1}$$

This exercise matches a published Stewart 8th Edition solution for Section 3.10 of *Early Transcendentals*. I haven't independently confirmed the numbering in Jerry's Section 2.9.

Exercise 17

Solution (a): Find the differential

Step 1: Recall the formula

$$\boxed{dy = f'(x)\,dx}$$

Step 2: Rewrite the function

$$y = (3 + x^2)^{1/2}$$

Step 3: Differentiate using the chain rule

$$\frac{dy}{dx} = \frac{1}{2}(3 + x^2)^{-1/2} \cdot (2x)$$

Simplify:

$$\frac{dy}{dx} = \frac{x}{\sqrt{3+x^2}}$$

Step 4: Multiply by dx

$$\boxed{dy = \frac{x}{\sqrt{3+x^2}}\,dx}$$

Solution (b): Evaluate dy

Given:

$$x = 1, \qquad dx = -0.1$$

Substitute:

$$dy = \frac{1}{\sqrt{3+1^2}}(-0.1)$$

$$= \frac{-0.1}{\sqrt{4}}$$

$$= \frac{-0.1}{2}$$

$$\boxed{dy = -0.05}$$

Final answers

(a)

$$\boxed{dy = \frac{x}{\sqrt{3+x^2}}\,dx}$$

(b)

$$\boxed{dy = -0.05}$$

Key concept for Jerry

Since $dx = -0.1$, the input x decreases slightly.

At $x = 1$, the function's slope is

$$f'(1) = \frac{1}{2}$$

Therefore, a change of -0.1 in x produces an approximate change of

$$dy \approx \frac{1}{2}(-0.1) = -0.05$$

Remember the distinction:

$$\boxed{dy \approx \Delta y}$$

The differential is a **linear approximation** to the actual change in the function.

You said:

problem 19 statement and solution

ChatGPT said:

Problem 19 — Page 193

James Stewart, Calculus, 8th Edition — Section 2.9: Linear Approximations and Differentials

Problem statement

For the function and values below, calculate both Δy and dy. Then sketch a diagram illustrating dx, dy, and Δy.

$$\boxed{y = x^2 - 4x, \quad x = 3, \quad dx = \Delta x = 0.5}$$

Verified against the textbook. Studylib

Solution

Step 1: Recall the formulas

The actual change in the function is:

$$\boxed{\Delta y = f(x + \Delta x) - f(x)}$$

The differential is:

$$\boxed{dy = f'(x)\,dx}$$

Step 2: Calculate Δy

Given:

$$f(x) = x^2 - 4x$$

First calculate $f(3)$:

$$f(3) = 3^2 - 4(3) = 9 - 12 = -3$$

Since $\Delta x = 0.5$, the new x-value is:

$$x + \Delta x = 3 + 0.5 = 3.5$$

Calculate $f(3.5)$:

$$f(3.5) = (3.5)^2 - 4(3.5)$$

$$= 12.25 - 14 = -1.75$$

Therefore:

$$\Delta y = f(3.5) - f(3)$$

$$= -1.75 - (-3)$$

$$\boxed{\Delta y = 1.25}$$

Step 3: Calculate dy

Differentiate:

$$f'(x) = 2x - 4$$

At $x = 3$:

$$f'(3) = 2(3) - 4 = 2$$

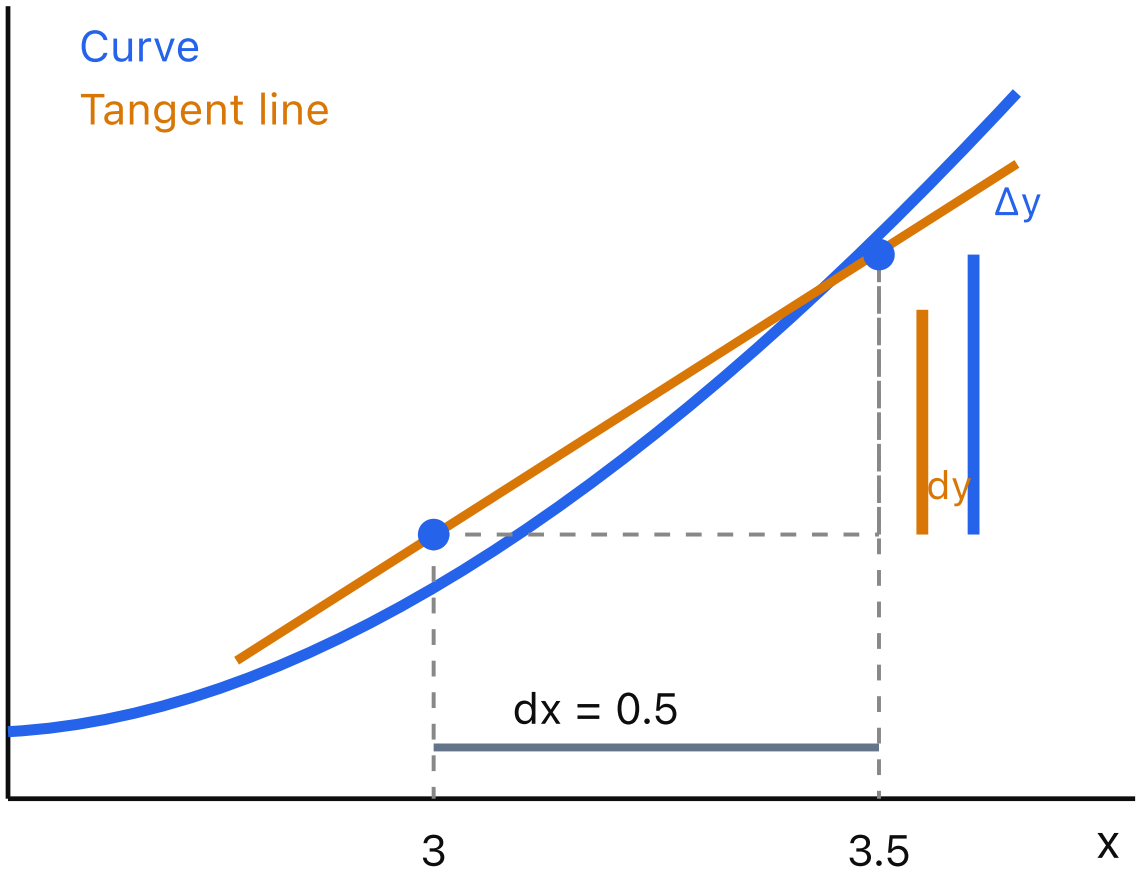
Using the differential formula:

$$dy = f'(3)\,dx$$

$$dy = 2(0.5)$$

$$\boxed{dy = 1}$$

Step 4: Understand the diagram



Conceptual sketch (not to scale). The curve rises more than the tangent line over this interval.

- $dx = 0.5$: horizontal change
- $dy = 1$: vertical change along the tangent line
- $\Delta y = 1.25$: actual vertical change along the curve

Final answers

$$\boxed{\Delta y = 1.25}$$

$$\boxed{dy = 1}$$

The difference is:

$$\Delta y - dy = 1.25 - 1 = 0.25$$

Key concept for Jerry

Why is Δy larger than dy?

Because $f(x) = x^2 - 4x$ curves upward, its tangent line at $x = 3$ lies below the curve nearby. Therefore, the tangent line underestimates the actual change.

Remember:

$$\boxed{\Delta y \approx dy}$$

They become closer as Δx approaches zero, but they are not generally equal.

You said:

problem 21 statement and solution

ChatGPT said:

Problem 21 — Page 193

James Stewart, Calculus, 8th Edition — Section 2.9: Linear Approximations and Differentials

Problem statement

Calculate Δy and dy for the given values of x and $dx = \Delta x$. Sketch a diagram illustrating these changes.

$$\boxed{y = \sqrt{x - 2}, \qquad x = 3, \qquad \Delta x = dx = 0.8}$$

This matches Exercise 21 in Stewart's 8th Edition *Early Transcendentals* solution reference, though I haven't independently confirmed the numbering in Jerry's edition. Exercise 21

Solution

Step 1: Recall the two formulas

Actual change in y:

$$\boxed{\Delta y = f(x + \Delta x) - f(x)}$$

Approximate change using the differential:

$$\boxed{dy = f'(x)\,dx}$$

Step 2: Calculate Δy

Given:

$$f(x) = \sqrt{x - 2}$$

First calculate:

$$f(3) = \sqrt{3 - 2} = 1$$

The new x-value is:

$$x + \Delta x = 3 + 0.8 = 3.8$$

Therefore:

$$f(3.8) = \sqrt{3.8 - 2} = \sqrt{1.8}$$

Calculate the actual change:

$$\Delta y = f(3.8) - f(3)$$

$$= \sqrt{1.8} - 1$$

$$\boxed{\Delta y \approx 0.34164}$$

Step 3: Calculate dy

Differentiate:

$$y = (x-2)^{1/2}$$

Using the chain rule:

$$f'(x) = \frac{1}{2}(x-2)^{-1/2}$$

$$\boxed{f'(x) = \frac{1}{2\sqrt{x-2}}}$$

Now substitute $x = 3$:

$$f'(3) = \frac{1}{2\sqrt{3-2}} = \frac{1}{2}$$

Calculate:

$$dy = f'(3)\,dx$$

$$= \frac{1}{2}(0.8)$$

$$\boxed{dy = 0.4}$$

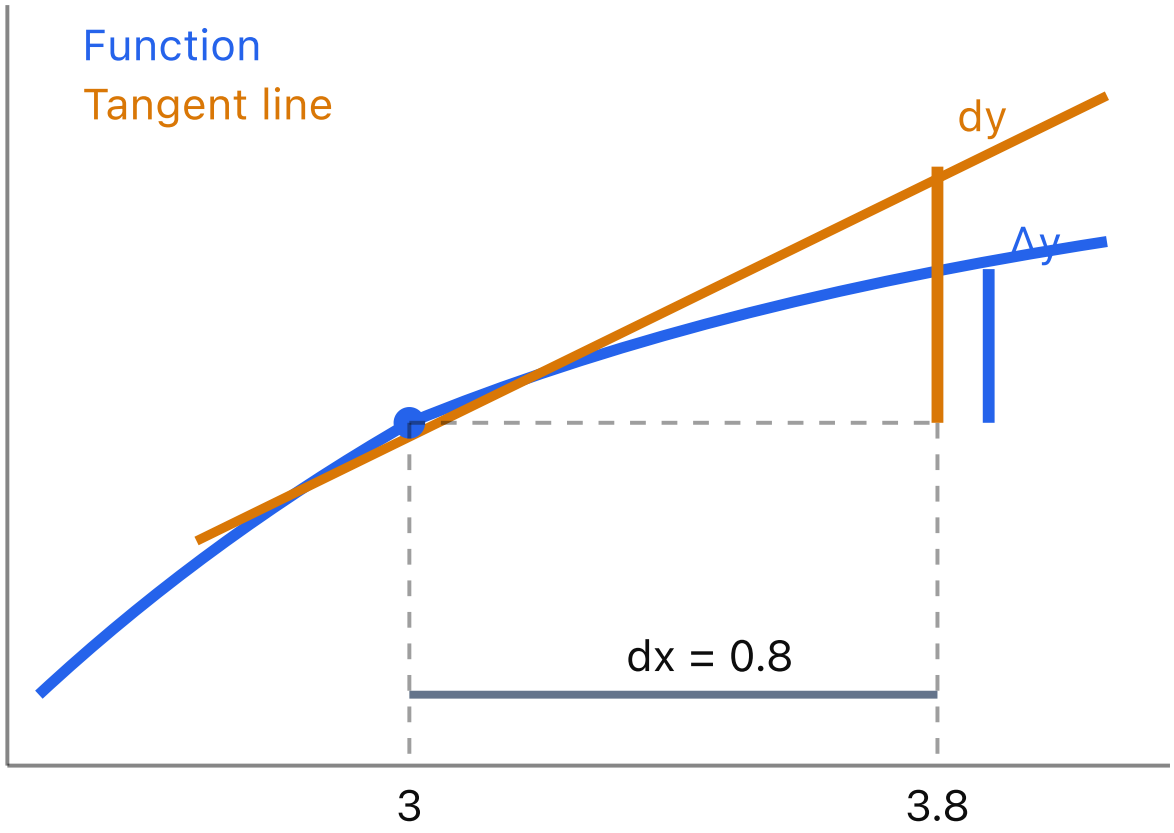
Step 4: Compare the two results

Actual change	Tangent-line estimate
0.34164	**0.4**
Δy	dy

The approximation error is:

$$dy - \Delta y = 0.4 - 0.34164 = \boxed{0.05836}$$

Step 5: Understand the graph



Conceptual illustration, not to scale. The tangent line rises faster than the square-root curve over the interval.

Final answers

$$\boxed{\Delta y = \sqrt{1.8} - 1 \approx 0.34164}$$

$$\boxed{dy = 0.4}$$

Key concept for Jerry

Notice that dy is **larger** than Δy, unlike problem #19.

Why?

Because $y = \sqrt{x-2}$ is **concave down**:

$$f''(x) = -\frac{1}{4(x-2)^{3/2}} < 0$$

The tangent line lies above the curve near $x = 3$. Therefore, linear approximation **overestimates** the actual change.

This is a useful AP Calculus concept:

- Concave up → tangent-line approximation tends to underestimate.
- Concave down → tangent-line approximation tends to overestimate.

You said:

problem 23 statement and solution

ChatGPT said:

Problem 23 — Page 193

James Stewart, Calculus, 8th Edition — Section 2.9: Linear Approximations and Differentials

Problem statement

Use a linear approximation (or differentials) to estimate:

$$\boxed{(1.999)^4}$$

Verified against the textbook solution manual. Test Bank Fan

Step 1: Choose a function and a nearby convenient point

Let:

$$f(x) = x^4$$

We want $f(1.999)$.

Since 1.999 is very close to 2, choose:

$$a = 2$$

Step 2: Recall the linearization formula

$$\boxed{L(x) = f(a) + f'(a)(x - a)}$$

Step 3: Calculate $f(a)$

$$f(2) = 2^4 = \boxed{16}$$

Step 4: Find the derivative

$$f'(x) = 4x^3$$

At $a = 2$:

$$f'(2) = 4(2)^3 = 4(8) = \boxed{32}$$

Step 5: Write the linearization

$$L(x) = 16 + 32(x - 2)$$

$$\boxed{L(x) = 16 + 32(x - 2)}$$

Step 6: Approximate $(1.999)^4$

Substitute $x = 1.999$:

$$(1.999)^4 \approx L(1.999)$$

$$= 16 + 32(1.999 - 2)$$

$$= 16 + 32(-0.001)$$

$$= 16 - 0.032$$

$$\boxed{(1.999)^4 \approx 15.968}$$

Final answer

$$\boxed{15.968}$$

Alternative: Using differentials

Remember:

$$dy = f'(x)\,dx$$

Choose $x = 2$, $dx = -0.001$:

$$dy = 4(2)^3(-0.001) = -0.032$$

Since

$$f(x + dx) \approx f(x) + dy$$

we obtain:

$$(1.999)^4 \approx 16 - 0.032 = \boxed{15.968}$$

Key concept for Jerry

The important skill is **choosing a nearby number that is easy to calculate**.

Here, $2^4 = 16$ is easy, and the adjustment from 2 to 1.999 is very small.

On an AP Calculus test, remember:

$$\boxed{f(a + \Delta x) \approx f(a) + f'(a)\Delta x}$$

You can estimate complicated numerical expressions using only a known function value, its derivative, and a small change in x.

You said:

problem 25 statement and solution

ChatGPT said:

Problem 25 — Page 193

James Stewart, Calculus, 8th Edition — Section 2.9: Linear Approximations and Differentials

Problem statement

Use a linear approximation (or differentials) to estimate:

$$\boxed{\sqrt[3]{1001}}$$

Verified against the textbook solution reference. Test Banks

Step 1: Choose a function and a convenient point

Let:

$$f(x) = \sqrt[3]{x} = x^{1/3}$$

We want to estimate $f(1001)$.

Since 1001 is close to 1000, choose:

$$a = 1000, \qquad \Delta x = 1$$

Because:

$$\sqrt[3]{1000} = 10$$

Step 2: Recall the linearization formula

$$\boxed{L(x) = f(a) + f'(a)(x - a)}$$

Step 3: Find the derivative

Using the power rule:

$$f'(x) = \frac{1}{3}x^{-2/3}$$

Rewrite:

$$f'(x) = \frac{1}{3x^{2/3}}$$

Evaluate at $a = 1000$:

$$f'(1000) = \frac{1}{3(1000)^{2/3}}$$

Since

$$(1000)^{2/3} = (\sqrt[3]{1000})^2 = 10^2 = 100$$

we get:

$$\boxed{f'(1000) = \frac{1}{300}}$$

Step 4: Write the linearization

$$L(x) = 10 + \frac{1}{300}(x - 1000)$$

$$\boxed{L(x) = 10 + \frac{x - 1000}{300}}$$

Step 5: Estimate $\sqrt[3]{1001}$

Substitute $x = 1001$:

$$\sqrt[3]{1001} \approx L(1001)$$

$$= 10 + \frac{1001 - 1000}{300}$$

$$= 10 + \frac{1}{300}$$

$$\boxed{\sqrt[3]{1001} \approx 10.00333}$$

Final answer

$$\boxed{\sqrt[3]{1001} \approx 10.00333}$$

Alternative solution using differentials

Recall:

$$dy = f'(x)\,dx$$

Choose $x = 1000$ and $dx = 1$:

$$dy = \frac{1}{3}(1000)^{-2/3}(1)$$

$$dy = \frac{1}{300}$$

Then:

$$f(1001) \approx f(1000) + dy$$

$$\boxed{\sqrt[3]{1001} \approx 10 + \frac{1}{300} = 10.00333}$$

Key concept for Jerry

The hardest part of this problem is recognizing that **1000 is a convenient nearby number because it is a perfect cube**.

Remember:

$$\boxed{f(a + \Delta x) \approx f(a) + f'(a)\Delta x}$$

In both problems #23 and #25, we follow exactly the same method: choose a nearby convenient number, calculate the derivative there, and use the small change to estimate the desired value.

You said:

problem 27 statement and solution

Worked for 29s

ChatGPT said:

Problem 27 — Page 193

James Stewart, Calculus, 8th Edition — Section 2.9: Linear Approximations and Differentials

Problem statement

Use a linear approximation (or differentials) to estimate:

$$\boxed{\tan 2^\circ}$$

Verified against the textbook solution reference. Test Banks

Step 1: Choose a function and a convenient point

Let:

$$f(x) = \tan x$$

We want to approximate $\tan 2^\circ$.

Since 2° is close to 0°, choose:

$$a = 0$$

Important: Calculus derivatives of trigonometric functions require angles in radians.

Convert:

$$2^\circ = \frac{2\pi}{180} = \frac{\pi}{90}$$

Step 2: Recall the linearization formula

$$\boxed{L(x) = f(a) + f'(a)(x - a)}$$

Step 3: Calculate $f(a)$

$$f(0) = \tan 0 = \boxed{0}$$

Step 4: Find the derivative

Recall:

$$f'(x) = \sec^2 x$$

Evaluate at $a = 0$:

$$f'(0) = \sec^2(0) = 1$$

Step 5: Write the linearization

$$L(x) = 0 + 1(x - 0)$$

$$\boxed{L(x) = x}$$

Therefore, for angles close to zero (in radians):

$$\boxed{\tan x \approx x}$$

Step 6: Approximate $\tan 2^\circ$

Substitute $x = \frac{\pi}{90}$:

$$\tan 2^\circ \approx L\left(\frac{\pi}{90}\right)$$

$$= \frac{\pi}{90}$$

$$\boxed{\tan 2^\circ \approx 0.03491}$$

Final answer

$$\boxed{\tan 2^\circ \approx \frac{\pi}{90} \approx 0.03491}$$

For comparison, the actual value is approximately 0.03492, so the approximation is very accurate.

Key concept for Jerry

This problem introduces an extremely useful small-angle approximation:

$$\boxed{\tan x \approx x \quad \text{when } x \approx 0}$$

The biggest potential mistake is **forgetting to convert degrees into radians**.

For example, writing $\tan 2^\circ \approx 2$ would be incorrect. We need:

$$2^\circ = \frac{\pi}{90} \text{ radians}$$

This same concept is important in physics, where small-angle approximations simplify many motion and oscillation problems.

You said:

problem 74 statement and solution

ChatGPT said:

Problem 74 — Page 198

James Stewart, Calculus, 8th Edition — Chapter 2 Review

Problem statement (paraphrased)

The volume of a right circular cone is

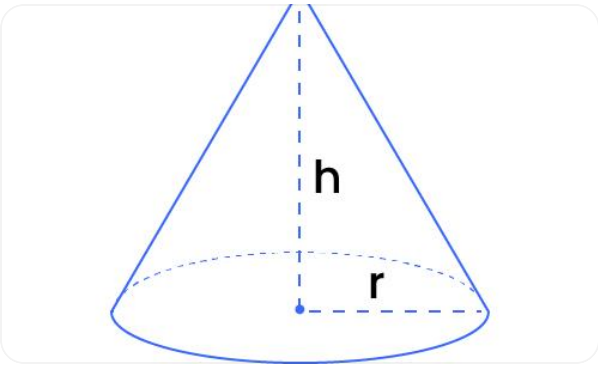
$$\boxed{V = \frac{1}{3}\pi r^2 h}$$

where r is the radius of the base and h is the height.

(a) Find $\dfrac{\partial V}{\partial h}$, treating r as constant.

(b) Find $\dfrac{\partial V}{\partial r}$, treating h as constant.

GradeSaver



Solution (a): Differentiate with respect to h

Start with:

$$V = \frac{1}{3}\pi r^2 h$$

Since r is constant, $\frac{1}{3}\pi r^2$ is also constant.

$$\frac{\partial V}{\partial h} = \frac{1}{3}\pi r^2 \frac{d}{dh}(h)$$

Therefore:

$$\boxed{\frac{\partial V}{\partial h} = \frac{1}{3}\pi r^2}$$

Solution (b): Differentiate with respect to r

Start with:

$$V = \frac{1}{3}\pi r^2 h$$

This time, h is constant.

$$\frac{\partial V}{\partial r} = \frac{1}{3}\pi h \frac{d}{dr}(r^2)$$

Using the power rule:

$$\frac{d}{dr}(r^2) = 2r$$

Therefore:

$$\frac{\partial V}{\partial r} = \frac{1}{3}\pi h(2r)$$

$$\boxed{\frac{\partial V}{\partial r} = \frac{2}{3}\pi r h}$$

Final answers

$$\boxed{\text{(a)} \quad \frac{\partial V}{\partial h} = \frac{1}{3}\pi r^2}$$

$$\boxed{\text{(b)} \quad \frac{\partial V}{\partial r} = \frac{2}{3}\pi r h}$$

Key concept for Jerry

The challenge is recognizing **which variable remains constant**.

- In (a), differentiate with respect to h, treating r as constant.
- In (b), differentiate with respect to r, treating h as constant.

Although this is essentially partial differentiation, the problem only requires the basic power rule and constant multiple rule.

You said:

problem 75 statement and solution

ChatGPT said:

Problem 75 — Page 198

James Stewart, Calculus, 8th Edition — Chapter 2 Review

Problem statement (paraphrased)

A rod has mass m, in kilograms, determined by its length x, in meters, according to

$$\boxed{m = x(1 + \sqrt{x})}$$

Find the **linear density** of the rod at $x = 4$ meters.

The answer is verified against the textbook solution reference. GradeSaver

Solution

Step 1: Understand linear density

Linear density describes how quickly mass increases with length.

$$\boxed{\rho(x) = \frac{dm}{dx}}$$

The unit is kilograms per meter (kg/m).

Step 2: Rewrite the mass function

Given:

$$m = x(1 + \sqrt{x})$$

Distribute x:

$$m = x + x\sqrt{x}$$

Since $\sqrt{x} = x^{1/2}$:

$$\boxed{m = x + x^{3/2}}$$

Step 3: Differentiate

Using the power rule:

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

Therefore:

$$\frac{dm}{dx} = 1 + \frac{3}{2}x^{1/2}$$

$$\boxed{\rho(x) = 1 + \frac{3}{2}\sqrt{x}}$$

Step 4: Evaluate at $x = 4$

$$\rho(4) = 1 + \frac{3}{2}\sqrt{4}$$

Since $\sqrt{4} = 2$:

$$= 1 + \frac{3}{2}(2)$$

$$= 1 + 3$$

$$\boxed{\rho(4) = 4 \text{ kg/m}}$$

Final answer

$$\boxed{\rho(4) = 4\ \text{kg/m}}$$

Alternative: Using the product rule

Jerry can also differentiate the original function directly:

$$m = x(1 + \sqrt{x})$$

Recall:

$$(fg)' = f'g + fg'$$

Therefore:

$$\frac{dm}{dx} = 1(1 + \sqrt{x}) + x\left(\frac{1}{2\sqrt{x}}\right)$$

Simplify:

$$= 1 + \sqrt{x} + \frac{\sqrt{x}}{2}$$

$$\boxed{\frac{dm}{dx} = 1 + \frac{3}{2}\sqrt{x}}$$

Same result!

Key concept for Jerry

The important skill is recognizing that **linear density is a derivative**.

Physical quantity	Calculus expression	Units
Mass	$m(x)$	kg
Length	x	m
Linear density	$\dfrac{dm}{dx}$	kg/m

At $x = 4$, the rod's linear density is $4\ \text{kg/m}$. This does **not** mean the entire rod has a mass of 4 kg. It means the mass is increasing at a rate of 4 kg per additional meter of length at that position.

Study tip: In physics applications, phrases such as *rate of change of mass with respect to length* tell you exactly which derivative to calculate.

You said:

problem 83 statement and solution

ChatGPT said:

Problem 83(a) — Page 198

James Stewart, Calculus, 8th Edition — Chapter 2 Review

Problem statement (paraphrased)

(a) Find the linear approximation of the function at $a = 0$, and use it to estimate $\sqrt[3]{1.03}$.

$$\boxed{f(x) = \sqrt[3]{1+3x}, \qquad a = 0}$$

Important: As Jerry's teacher clarified, the expression inside the cube root is $1 + 3x$, not $1 + x$. Test Bank Fan

Step 1: Recall the linearization formula

$$\boxed{L(x) = f(a) + f'(a)(x - a)}$$

Step 2: Calculate $f(a)$

Since $a = 0$:

$$f(0) = \sqrt[3]{1 + 3(0)} = 1$$

$$\boxed{f(0) = 1}$$

Step 3: Find the derivative

Rewrite the function:

$$f(x) = (1 + 3x)^{1/3}$$

Apply the power rule and chain rule:

$$f'(x) = \frac{1}{3}(1 + 3x)^{-2/3} \cdot 3$$

The $\frac{1}{3}$ and 3 cancel:

$$\boxed{f'(x) = (1 + 3x)^{-2/3}}$$

At $x = 0$:

$$f'(0) = (1 + 0)^{-2/3} = 1$$

$$\boxed{f'(0) = 1}$$

Step 4: Find the linearization

$$L(x) = 1 + 1(x - 0)$$

$$\boxed{L(x) = 1 + x}$$

Step 5: Approximate $\sqrt[3]{1.03}$

We need:

$$1 + 3x = 1.03$$

Solve for x:

$$3x = 0.03$$

$$\boxed{x = 0.01}$$

Now substitute into the linearization:

$$\sqrt[3]{1.03} \approx L(0.01)$$

$$= 1 + 0.01$$

$$\boxed{\sqrt[3]{1.03} \approx 1.01}$$

Final answer

$$\boxed{L(x) = 1 + x}$$

$$\boxed{\sqrt[3]{1.03} \approx 1.01}$$

Key concept for Jerry

The most important part is applying the **chain rule correctly**:

$$\frac{d}{dx}(1+3x)^{1/3} = \frac{1}{3}(1+3x)^{-2/3} \cdot 3$$

The extra factor of 3 changes the derivative and the tangent-line approximation.

Also, when approximating $\sqrt[3]{1.03}$, we must use $x = 0.01$, **not** $x = 0.03$.

This is similar to problem #6, but the $3x$ inside the cube root makes a crucial difference.

You said:

now put the problem statement and solution (step by step with key concept) for each problem we went though into a md file

Worked for 2m 8s

ChatGPT said:

I've compiled all **17 problems from Jerry's AP Calculus Homework #34** into one Markdown file.

Download HW #34 — Complete Worked Solutions (.md) ↗
17 problems · 372 lines · Markdown with LaTeX equations

Each problem includes its paraphrased statement, step-by-step solution, boxed final answer, and a **Key Concept for Jerry** explaining the important calculus principle or common mistake.

The file also includes a quick answer index, a formula review section, and the teacher's correction to **#83(a):** $1 + 3x$ inside the cube root.

Note: The solutions reflect our previous discussions. Some exercise statements still need verification against the printed textbook.

You said:

could you include the diagram for problem 21?

Worked for 45s

ChatGPT said:

Yes — I updated the Markdown file to include the **Problem 21 diagram**.

If you want, I can also:

1. add the **Problem 19 diagram** too, or
2. convert this updated Markdown into a **PDF**.

Jerry_AP_Calculus_HW34_Complete_Worked_Solutio...
Document

You said:

svg does not render inside github, if there a way to render it?

ChatGPT said:

Yes. **GitHub can display SVG diagrams in Markdown, but it does not render SVG code inside a fenced code block** such as ```` ```svg ```` .

The best solution is to save the diagram as a separate `.svg` file and reference it from Markdown:

```
![Problem 21 — Linear Approximation](problem21.svg)
```

GitHub will display the diagram as an image, provided the SVG file is committed to the repository at that relative path.

Alternatively, use a PNG image, which offers the most consistent compatibility across Markdown viewers.

I'll update Jerry's Homework #34 Markdown file to use an image reference instead of embedded SVG code.

Updated GitHub-ready files

I've replaced the embedded SVG code with a PNG reference in Jerry's Homework #34 guide.

Updated Homework #34 Markdown ↗

Problem 21 Diagram — PNG ↗

Problem 21 Diagram — SVG ↗

The diagram is available in the downloadable PNG file above.

To display it on GitHub: Upload the Markdown file and `problem21_linear_approximation.png` into the **same repository folder**. The diagram will automatically render inside Problem 21.

You can use the SVG instead if you prefer scalable graphics; simply change the image filename from `.png` to `.svg` in the Markdown.